Optimal Interpolation Formulas Exact for Trigonometric Functions

Samandar S.Babaev,[1, 2, a)] Javlon R. Davronov,[2, b)] Abdulla Abdullaev,[3] and Sarvarbek Z.Polvonov[1]

[1)] *Bukhara State University, 11, M.Ikbol str., Bukhara 200114, Uzbekistan.*
[2)] *V.I.Romanovskiy Institute of Mathematics, Uzbekistan Academy of Sciences, 46, University str., Tashkent 100174, Uzbekistan.*
[3)] *National University of Uzbekistan named after M.Ulugbek, Tashkent,Uzbekistan.*

[a)] *Corresponding author:bssamandar@gmail.com*
[b)] *Electronic mail: javlondavronov77@gmail.com*

Abstract.
In the present paper we investigate the problem of construction of the optimal interpolation formulas in the space $K_2(P_2)$. We find the norm of the error functional which gives the upper bound for the error of the interpolation formulas in the space $K_2(P_2)$. Further, we get the system of linear equations for coefficients of the optimal interpolation formulas. Finally, using the discrete analogue of the differential operator $\frac{d^4}{dx^4} + 2\omega^2 \frac{d^2}{dx^2} + \omega^4$ and its properties we find explicit formulas for the coefficients of the optimal interpolation formulas.

INTRODUCTION

There are algebraic and variational approaches in the spline theory. In the algebraic approach splines are considered as some smooth piecewise polynomial functions. In the variational approach splines are elements of Hilbert or Banach spaces minimizing certain functionals. The first spline functions were constructed from pieces of cubic polynomials. After that, this construction was modified, the degree of polynomials increased. The theory of splines based on variational methods were studied and developed, for example, by S.L.Sobolev [1], C. de Boor [2], I.J.Schoenberg [3], J.H.Ahlberg et al. [4], L.L.Schumaker [5], V.A.Vasilenko [6], G.V.Milovanivić, Kh.M.Shadimetov and A.R.Hayotov [7], S.S.Babaev [8, 9, 10] and others. The present work is also devoted to the variational method of construction of optimal interpolation formulas.

We consider the following interpolation formula

$$\varphi(x) \cong P_\varphi(x) = \sum_{\beta=0}^{N} C_\beta(x)\varphi(x_\beta). \tag{1}$$

Here C_β and $x_\beta (\in [0,1])$ are the *coefficients* and the *nodes* of the interpolation formula, respectively. We suppose that functions φ belong to the Hilbert space

$$K_2(P_2) := \left\{ \varphi : [0,1] \to \mathbb{R} | \varphi' \text{ is abs.cont. and } \varphi'' \in L_2(0,1) \right\}$$

equipped with the norm

$$\|\varphi\|_{K_2} = \left\{ \int_0^1 \left(\varphi''(x) + \omega^2\varphi(x)\right)^2 dx \right\}^{1/2},$$

where $\omega \in \mathbb{R}$ and $\omega \neq 0$. The inner product of functions φ and ψ in the space $K_2(P_2)$ is defined as

$$\langle \varphi, \psi \rangle_{K_2} = \int_0^1 \left(\varphi''(x) + \omega^2\varphi(x)\right) \cdot \left(\psi''(x) + \omega^2\psi(x)\right) dx.$$

The error of the interpolation formula (1) at a point $z \in [0,1]$ is the following difference

$$(\ell, \varphi) = \varphi(z) - \sum_{\beta=0}^{N} C_\beta(z)\varphi(x_\beta)$$

which is the value of a functional ℓ at a function φ. Here the functional ℓ is defined as

$$\ell(x,z) = \delta(x-z) - \sum_{\beta=0}^{N} C_\beta(z)\delta(x-x_\beta) \tag{2}$$

and it is called *the error functional*. Here δ is Dirac's delta-function.

According to the Riesz theorem any linear continuous functional ℓ on the Hilbert space is represented in the form of a inner product. Therefore, in our case, for any function φ from the space $K_2(P_2)$, there is a function $\psi_\ell \in K_2(P_2)$ satisfying the equality

$$(\ell, \varphi) = \langle \varphi, \psi_\ell \rangle_{K_2} \tag{3}$$

and

$$\|\ell\|_{K_2^*} = \|\psi_\ell\|_{K_2}.$$

It should be noted that according to the Cauchy-Schwarz inequality the absolute value of the error of the formula (1) is estimated by the norm of the error functional as follows

$$|(\ell, \varphi)| \le \|\ell\|_{K_2^*} \cdot \|\varphi\|_{K_2}.$$

We note that the function ψ_ℓ satisfying the equality in the last inequality is called *the extremal function* for the functional ℓ (see, [11]) .

Consequently, **the problem** of constructing the optimal interpolation formula in the space $K_2(P_2)$ is to found the following quantity for fixed nodes x_β:

$$\|\mathring{\ell}\|_{K_2} = \inf_{C_\beta(z)} \sup_{\|\varphi\|_{K_2}=1} |(\ell, \varphi)|.$$

The coefficients satisfying the last equality are called *the optimal coefficients* and they are denoted as $\mathring{C}_\beta$ (if exist). The interpolation formula (1) with coefficients $\mathring{C}_\beta$ is called *the optimal interpolation* formula and $\mathring{\ell}$ is the error functional corresponding to the optimal interpolation formula (1).

THE EXTREMAL FUNCTION AND THE NORM OF THE ERROR FUNCTIONAL

We racall that the function $\psi_\ell(x)$, which satisfies the following equality

$$(\ell, \psi_\ell) = \|\ell\|_{K_2^*} \cdot \|\psi_\ell\|_{K_2},$$

is called the extremal function of the formula (1), see, for example [1, 11, 12].

Using the integration by parts for the inner product $\langle \psi_\ell, \varphi \rangle_{K_2(P_2)}$ we have

$$\langle \psi_\ell, \varphi \rangle_{K_2} = \sum_{i=0}^{1} (-1)^i (\psi_\ell^{(i+2)}(x) + \omega^2 \psi_\ell^{(i)}(x)) \varphi^{(i)}(x)|_0^1 + \int_0^1 \varphi(x) \cdot \left(\psi_\ell^{(4)} + 2\omega^2 \psi_\ell''(x) + \omega^4 \psi_\ell(x)\right) \mathrm{dx}.$$

Hence, for ψ_ℓ we come to the following differential equation

$$\psi_\ell^{(4)} + 2\omega^2 \psi_\ell''(x) + \omega^4 \psi_\ell(x) = \ell(x) \tag{4}$$

with the boundary conditions

$$\psi_\ell^{(i+2)}(x) + \omega^2 \psi_\ell^{(i)}(x)|_0^1 = 0,\ i = 0, 1. \tag{5}$$

The solution of the boundary value problem (4)-(5) has the following form

$$\psi_\ell(x) = (-1)^2 \ell(x) * G_2(x) + d_1 \sin(\omega x) + d_2 \cos(\omega x), \tag{6}$$

where

$$G_2(x) = \frac{\operatorname{sign}(x)}{4 \cdot \omega^3} \cdot \Big(\sin(\omega x) - \omega x \cdot \cos(\omega x) \Big).$$

By the Riesz theorem for the square of the norm of the error functional the following equality holds $\|\ell\|^2_{\tilde{K}_2^*(P_2)} = (\ell, \psi_\ell)$. Hence, using (2) and (6) for the norm of the error functional we get the following expression

$$\|\ell\|^2_{K_2} = \sum_{\beta=0}^{N} \sum_{\gamma=0}^{N} C_\beta(z) C_\gamma(z) G_2(x_\beta - x_\gamma) - 2 \sum_{\beta=0}^{N} C_\beta(z) G_2(z - x_\beta). \tag{7}$$

Thus, using the last equality we can get an upper bound for the error of the interpolation formulas (1). Further, in order to obtain the optimal interpolation formula of the form (1) we should find the minimum of the expression (7) by coefficients $C_\beta(z)$ under the conditions $\sum\limits_{\beta=0}^{N} C_\beta(z) \cdot \sin(\omega h \beta) = \sin(\omega z)$ and $\sum\limits_{\beta=0}^{N} C_\beta(z) \cdot \cos(\omega h \beta) = \cos(\omega z)$. For this we apply the Lagrange method.

Consider the function

$$\Psi(C, d_1, d_2) = \|\ell\|^2 + d_1(\ell, \sin(\omega x)) + d_2(\ell, \cos(\omega x))$$

where $C = (C_0, C_1, C_2, \ldots, C_N)$, d_1 and d_2 are Lagrange multipliers.

Equating to 0 the partial derivatives of the function Ψ by C_β, $\beta = \overline{0,N}$, d_1 and d_2, we get the following system of linear equations

$$\sum_{\gamma=0}^{N} C_\gamma(z) G_2(x_\beta - x_\gamma) + d_1 \sin(\omega h \beta) + d_2 \cos(\omega h \beta) = G_2(z - x_\beta), \ \beta = \overline{0.N}, \tag{8}$$

$$\sum_{\beta=0}^{N} C_\beta(z) \sin(\omega h \beta) = \sin(\omega z), \tag{9}$$

$$\sum_{\beta=0}^{N} C_\beta(z) \cos(\omega h \beta) = \cos(\omega z). \tag{10}$$

We have obtained the linear system of $N+3$ unknowns with $N+3$ equations. Next, we solve the system (8)-(10).

AN ALGORITHM FOR CALCULATING THE COEFFICIENTS OF THE OPTIMAL INTERPOLATION FORMULA

In this section, we use the concept of a function with a discrete argument and the operations performed on them [11, 12, 13].

The function $\varphi(h\beta)$ is a *discrete argument function* (or *discrete function*) if it is given on some set of integer values of β.

Suppose that $C_\beta = 0$ when $\beta < 0$ and $\beta > N$, $x_\beta = h\beta$, $h = \frac{1}{N}$. We rewrite the system (8)-(10) in the convolution form

$$\begin{cases} C_\beta(z) * G_2(h\beta) + d_1 \cdot \sin(\omega h \beta) + d_2 \cdot \cos(\omega h \beta) = G_2(z - h\beta), \ \ \beta = \overline{0.N}, \\ \sum\limits_{\beta=0}^{N} C_\beta(z) \cdot \sin(\omega h \beta) = \sin(\omega z), \\ \sum\limits_{\beta=0}^{N} C_\beta(z) \cdot \cos(\omega h \beta) = \cos(\omega z), \\ C_\beta(z) = 0, \qquad\qquad \beta < 0 \text{ and } \beta > N. \end{cases} \tag{11}$$

We have the following

Problem 1. Find the discrete function $C_\beta(z)$ and constants d_1, d_2 which satisfy the system (11).

It should be noted that if we solve Problem 1 we get the optimal coefficients $\mathring{C}_\beta$, $\beta = 0,1,\ldots,N$. We do not solve the system (11) by a direct method. Instead we give an algorithm which is used a discrete analog of the differential operator $\frac{d^4}{dx^4} + 2\omega^2\frac{d^2}{dx^2} + \omega^4$. This algorithm allows us to get the explicit formulas for coefficients of the optimal interpolation formula (1).

Further, we explain the algorithm.

We consider the following two functions

$$v(h\beta) = C_\beta(z) * G_2(h\beta) = \sum_{\gamma=0}^{N} C_\gamma(z) G_2(h\beta - h\gamma)$$

and

$$u(h\beta) = v(h\beta) + d_1\sin(\omega h\beta) + d_2\cos(\omega h\beta).$$

Now we should express the coefficients $C_\beta(z)$ by the function $u(h\beta)$. For this we use the discrete analog $D_2[\beta]$ of the operator $\frac{d^4}{dx^4} + 2\omega^2\frac{d^2}{dx^2} + \omega^4$. The discrete argument function $D_2[\beta]$ satisfies the following equation

$$D_2[\beta] * G_2[\beta] = \delta[\beta] \tag{12}$$

where $\delta[\beta]$ is equal to 0 when $\beta \neq 0$ and is equal to 1 when $\beta = 0$, i.e. $\delta[\beta]$ is the discrete delta-function.

Theorem 2. The discrete analogue $D_2[\beta]$ of the differential operator $\frac{d^4}{dx^4} + 2\omega^2\frac{d^2}{dx^2} + \omega^4$ satisfying (12) has the form

$$D_2[\beta] = K \cdot \begin{cases} A_1 \cdot \lambda_1^{|\beta|-1}, & |\beta| \geq 2, \\ 1 + A_1, & |\beta| = 1, \\ Y + \frac{A_1}{\lambda_1}, & \beta = 0, \end{cases}$$

where $\omega \in \mathbb{R}$, $\omega \neq 0$ and A_1,K,Y,λ_1 are know.

Theorem 2 was proved on the work [14].

Then for the optimal coefficients $C_\beta(z)$ we have

$$C_\beta(z) = D_2[\beta] * u[\beta]. \tag{13}$$

Thus, if we find the function $u(h\beta)$ then the optimal coefficients $\mathring{C}_\beta(z)$ will be found from equality (13). In order to calculate the convolution in (13) it is required to find the representation of the function $u(h\beta)$ for all integer values of β. From equality (11) we get that $u(h\beta) = G(z - h\beta)$ for $h\beta \in [0,1]$. Now we find the representation of the function $u(h\beta)$ for $\beta < 0$ and $\beta > N$.

Since $C_\beta = 0$ when $h\beta \notin [0,1]$ then we get

$$C_\beta(z) = D_2[\beta] * u[\beta] = 0\,,\ h\beta \notin [0,1].$$

Further, we calculate the convolution $v(h\beta) = G_2(h\beta) * C_\beta$ for $h\beta \notin [0;1]$. Suppose that $\beta \leq 0$ then

$$v(h\beta) = -\left[b_1 \cdot \sin(\omega h\beta) + b_2 \cdot \cos(\omega h\beta) - \frac{1}{4\omega^3} \cdot (\omega h\beta) \cdot \cos(\omega z - \omega h\beta)\right].$$

Similarly, in the case $\beta \geq N$ for the convolution $v(h\beta) = G_2(h\beta) * C_\beta(z)$ we obtain

$$v(h\beta) = b_1 \cdot \sin(\omega h\beta) + b_2 \cdot \cos(\omega h\beta) - \frac{1}{4\omega^3} \cdot (\omega h\beta) \cdot \cos(\omega z - \omega h\beta)$$

where $b_1 = \frac{1}{4\omega^3} \cdot \Big(\cos(\omega z) + \sum\limits_{\gamma=0}^{N} C_\gamma(z)(\omega h\gamma)\sin(\omega h\gamma)\Big)$, $b_2 = \frac{1}{4\omega^3} \cdot \Big(\sum\limits_{\gamma=0}^{N} C_\gamma(z)(\omega h\gamma)\cos(\omega h\gamma) - \sin(\omega z)\Big)$ according to this

$$C_\beta(z) = D_2[\beta] * u[\beta] \tag{14}$$

here

$$u(h\beta) = v(h\beta) + d_1 \sin(\omega h\beta) + d_2 \cos(\omega h\beta),$$
$$v(h\beta) = C_\beta(z) * G_2(h\beta) = \sum_{\gamma=0}^{N} C_\gamma(z) G_2(h\beta - h\gamma),$$

As we know, $C_\beta(z) = 0$ when $\beta < 0$ and $\beta > N$. Then we have

$$u[\beta] = \begin{cases} a_1^- \cdot \sin(\omega h\beta) + a_2^- \cos(\omega h\beta) + \frac{1}{4\omega^3}(\omega h\beta) \cdot \cos(\omega z - \omega h\beta), & \beta < 0, \\ G_2(z - h\beta), & 0 \le \beta \le N, \\ a_1^+ \cdot \sin(\omega h\beta) + a_2^+ \cos(\omega h\beta) - \frac{1}{4\omega^3}(\omega h\beta) \cdot \cos(\omega z - \omega h\beta), & \beta > N, \end{cases} \tag{15}$$

where $a_1^- = d_1 - b_1$, $a_2^- = d_2 - b_2$, $a_1^+ = d_1 + b_1$, $a_2^+ = d_2 + b_2$ are denoted. In (15) we put $\beta = 0$ and $\beta = N$ and we come

$$\begin{cases} a_2^- = G_2(z), \\ a_1^+ \cdot \sin(\omega) + a_2^+ \cdot \cos(\omega) - \frac{1}{4\omega^2} \cdot \cos(\omega z - \omega) = G_2(z-1), \end{cases}$$

from this we find a_2^- and a_2^+.

$$\begin{cases} a_2^- = \frac{1}{4\omega^3}\Big(\sin(\omega z) - \omega z \cdot \cos(\omega z)\Big), \\ a_2^+ = -\frac{\sin(\omega z - \omega) - \omega z \cdot \cos(\omega z - \omega)}{4\omega^3 \cdot \cos(\omega)} - a_1^+ \tan(\omega), \end{cases} \tag{16}$$

now we use $C_\beta(z) = 0$ in $\beta < 0$ and $\beta > N$ to find the remaining unknowns, i.e.

$$C_\beta(z) = D_2(h\beta) * u(h\beta) = 0$$

then we calculate the last equation for the coefficients $C_\beta(z)$ and we have the following.

$$C_\beta(z) = \sum_{\gamma=1}^{+\infty} D_2(h\beta + h\gamma) \cdot \left[G_2(z + h\gamma) - \sin(\omega h\gamma) \cdot \Big(a_1^- + \frac{\cos(\omega z) + \omega z \sin(\omega z)}{4\omega^3} \Big) \right]$$

$$+ \sum_{\gamma=0}^{N} D_2(h\beta - h\gamma) G_2(z - h\gamma) + \sum_{\gamma=1}^{+\infty} D_2(h(\gamma + N - \beta)) \cdot \left[G_2(z - 1 - h\gamma) + \sin(\omega h\gamma) \Big(\frac{a_1^+}{\cos(\omega)} - \frac{\cos(\omega z) + \omega z \sin(\omega z)}{4\omega^3 \cos(\omega)} \Big) \right].$$

Substituting $\beta = -1$ and $\beta = N + 1$, we get a system of linear equations with respect to a_1^- and a_1^+

$$\begin{cases} -a_1^- M_{11} + \frac{a_1^+}{\cos(\omega)} M_{12} = T_1, \\ -a_1^- M_{12} + \frac{a_1^+}{\cos(\omega)} M_{11} = T_2, \end{cases} \tag{17}$$

from the system of equations above we find a_1^- and a_1^+

$$\begin{cases} a_1^- = \frac{M_{12} T_2 - M_{11} T_1}{M_{11}^2 - M_{12}^2}, \\ a_1^+ = \frac{M_{11} T_2 - M_{12} T_1}{M_{11}^2 - M_{12}^2} \cdot \cos(\omega), \end{cases} \tag{18}$$

where T_1 T_2 M_{11} M_{12} are known.

Then solving the systems (16), (18) of equations we get a_1^-, a_1^+, a_2^- and a_2^+. Finally, from (14) for $\beta = 0, 1, \ldots, N$ we get the explicit formulas for optimal coefficients as claimed in the following theorem.

Theorem 3. *Coefficients of the optimal interpolation formula (1) with equal spaced nodes in the space $K_2(P_2)$ have the following from*

$$\mathring{C}_0(z) = K\cdot\Bigg[G_2(z+h) - \Big(a_1^- + \frac{\cos(\omega z)+\omega z\sin(\omega)}{4\omega^3}\Big)\cdot\sin(\omega h\gamma) + YG_2(z) + G_2(z-h)$$
$$+ \frac{A_1}{\lambda_1}\Big[\sum_{\gamma=0}^{N}\lambda_1^{\gamma}G_2(z-h\gamma) + \lambda_1^N N_1 + M_1\Big]\Bigg],$$

$$\mathring{C}_\beta(z) = K\cdot\Bigg[YG_2(z-h\beta) + G_2(z-h(\beta-1)) + G_2(z-h(\beta+1)) + \frac{A_1}{\lambda_1}\Big[\sum_{\gamma=0}^{N}\lambda_1^{|\beta-\gamma|}G_2(z-h\gamma) + \lambda_1^{N-\beta}N_1 + \lambda_1^{\beta}M_1\Big]\Bigg],$$

$$\mathring{C}_N(z) = K\cdot\Bigg[YG_2(z-1) + G_2(z-1+h) + G_2(z-1-h) + \frac{A_1}{\lambda_1}\Big[\sum_{\gamma=0}^{N}\lambda_1^{N-\gamma}G_2(z-h\gamma) + N_1 + \lambda_1^N M_1\Big]$$
$$+ \Big(\frac{a_1^+}{\cos(\omega)} - \frac{\cos(\omega z)+\omega z\sin(\omega z)}{4\omega^3\cos(\omega)}\Big)\sin(\omega h)\Bigg].$$

CONCLUSION

In this work, the extremal function of the error functional of the optimal interpolation formula (1) was determined in $K_2(P_2)$ space. Also, we find explicit form of coefficients of the optimal interpolation formula (1).

ACKNOWLEDGMENTS

We would like to thank our colleagues at Bukhara State University and the Institute of Mathematics named V.I.Romanovskiy for making convenient research facilities.

REFERENCES

1. S. L. Sobolev, "On interpolation of functions of *n* variables, in: Selected works of S.L. Sobolev," Springer , 451–456 (2006).
2. C. de Boor, "Best approximation properties of spline functions of odd degree," J. Math. Mech. **12**, 747–749 (1963).
3. I. J. Schoenberg, "On trigonometric spline interpolation," J. Math. Mech. 13 , 795–825 (1964).
4. J. H. Ahlberg, E. N. Nilson, and J. L. Walsh, *The theory of splines and their applications, Mathematics in Science and Engineering* (New York: Academic Press, 1967).
5. L. Schumaker, "Spline functions:basic theory," Cambridge university press , 600 (2007).
6. V. Vasilenko, "Spline-fucntions: Theory, algorithms, programs,," Novosibirsk , 216 (1983).
7. A. R. Hayotov, G. V. Milovanović, and Kh. M. Shadimetov, "Optimal quadrature formulas and interpolation splines minimizing the semi-norm in $K_2(P_2)$ space," Milovanović.G.V and Rassias.M.Th (eds.), Analytic Number Theory, Approximation Theory, and Special Functions, Springer, , 573–611 (2014).
8. S. S. Babaev, J. R. Davronov, and N. H. Mamatova, "On an optimal interpolation formula in the space $W_{2,\sigma}^{(1,0)}$," Bulletin of the Institute of Mathematics , 1–12 (2020).
9. S. S. Babaev and A. R. Hayotov, "Optimal interpolation formulas in the space $W_2^{(m,m-1)}$," Calcolo **56**, doi.org/10.1007/s10092–019–0320–9 (2019).
10. A. R. Hayotov and S. S. Babaev, "Optimal quadrature formulas for computing of Fourier integrals in $W_2^{(m,m-1)}$ space," AIP Conference Proceedings **2365**, 020021 (2021), https://doi.org/10.1063/5.0057127.
11. S. L. Sobolev, *Introduction to the Theory of Cubature Formulas* (Moscow, 1974) p. 808.
12. S.L.Sobolev, "The coefficients of optimal quadrature formulas, in: Selected works of S.L.Sobolev," Springer , 561–566 (2006).
13. S. L. Sobolev and V. L. Vaskevich, *The Theory of Cubature Formulas.* (1997).
14. A. R. Hayotov, "Discrete analogues of some differential operators," Uzbek Mathematical Journal , 151–155 (2012).